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EFFECTS OF MATERNAL AND PEER SEPARATIONS ON YOUNG MONKEYS

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DURING the past decade numerous investigators have closely examined the behavioral reactions of young monkeys to physical separation from social attachment objects. Most studies have focused upon the consequences of maternal separation in rhesus macaques during their first 6 months of life. It has consistently been found that the infants respond to maternal separation with active protest and sometimes with subsequent depression and that during separation they exhibit marked decreases in levels of complex activities such as play (Seay, Hansen and Harlow, 1962; Seay and Harlow, 1965; Hinde, Spencer-Booth and Bruce, 1966; Spencer-Booth and Hinde, 1971; Abrams, 1970). Most subjects apparently return to preseparation baselines of activity following reunion with mother, although residual effects of the separations, e.g. extreme sensitivity to fear stimuli, have been found in some monkeys many months after reunion (Mitchell, Harlow, Griffin and Møller, 1967; Hinde and Davies, 1972; Suomi, Collins and Harlow, 1973; Young, Suomi, Harlow and McKinney, 1973). Several reviewers (Kaufman and Rosenblum, 1967; McKinney and Bunney, 1969; Harlow, Suomi and McKinney, 1970; Hinde and Spencer-Booth, 1971; Harlow and Suomi, 1974) have pointed out the similarity between these data and those reported for human anaclitic depression precipitated by separation from mother.

However, other researchers have examined the reactions of different monkey species to maternal separations and their findings have not always paralleled those from rhesus monkeys and human infant separation studies. For example, Kaufman and Rosenblum (1969) found that pigtail macaque infants essentially mirrored the rhesus response to maternal separation but bonnet macaque infants were scarcely affected by identical manipulations. Similarly, Seay and his colleagues (Preston, Baker and Seay, 1970; Schlottmann and Seay, 1972) reported different patterns of separation reaction for patas monkeys and Java monkeys, the former exhibiting primarily a change in type of play pattern in response to removal of mother, the latter reacting more in the fashion of rhesus and pigtail infants. Clearly, considerable species variation exists with respect to separation reaction, a fact which most investigators have attributed to species variation in strength and specificity of mother–infant attachment prior to separation.

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More recent data have disclosed that even among rhesus monkeys reactions to maternal separation are far from universal across individuals or situations (Suomi, in press; Young, McKinney and Lewis, in press). Furthermore, it has been demonstrated that mother–infant bonds are not the only simian relationships strong enough to produce adverse reaction to separation. Infants reared with and subsequently separated from peers (Suomi, Harlow and Domek, 1970) and/or surrogate mothers (Novak, 1973) have exhibited reactions as severe and debilitating as any seen following rhesus mother–infant separation.

In view of these facts, recent work in the study of social separation has been directed toward uncovering factors which determine and variables which predict the course and consequence of a given separation of a particular infant from any object of attachment. It has been found, for example that age of the subject at time of separation, form and intensity of attachment relationship and nature of social environment prior to separation, type and duration of social separation and social structure of reunion environment are variables which can profoundly influence the reaction of a young monkey to loss of an attachment object. These findings have been systematically discussed in reviews by Hinde and Spencer-Booth (1971), McKinney, Suomi and Harlow (in press) and Suomi (in press).

The following experimentation represents a continuation of research designed to examine factors which influence response to social separation in rhesus monkeys. Young monkeys were subjected to three consecutive 2-week separations, each producing an increasing degree of social isolation. The infants were first separated from their mothers by partitions which permitted limited physical contact through wide-gauge wire mesh and unlimited interaction with familiar peers. They were then denied all physical contact with their mothers by replacement of the mesh partitions with transparent plastic panels, and finally were separated from their familiar peers as well. Their behaviors were compared with those of a group of like-aged monkeys who had not been subjected to separations of any form. On the basis of previous data it was predicted that the experimental subjects would exhibit increasingly severe reactions to each separation and that clear differences in levels of behaviors of experimental and control monkeys would be found, even long after reunion.

METHOD

Subjects and procedure

The subjects were 8 rhesus monkeys (*Macaca mulatta*) assigned at birth to one of two groups, separation (*S*) or control (*C*). The *S* group contained 1 male and 3 females, all born within a 24-day period. The *C* group consisted of 2 males and 2 females born within a 15-day period. Immediately after birth each subject was housed with its mother in one of the living cages of a playpen apparatus (Hansen, 1966). This apparatus, illustrated in Fig. 1, was a living-experimental unit consisting of four 94 × 94 × 94 cm living cages surrounding a 152 × 152 × 76 cm play area. The 9 × 14 cm openings in each living cage permitted infant access to the play area but were too small to allow mothers to leave their respective living cages. The play area could be subdivided into four 76 × 76 × 76 cm compartments by inserting 76 × 76 cm partitions into formed slots. All subjects in each group were housed in the same playpen, so that beginning shortly after birth, each subject had unlimited access to not only its mother but also the other infants in its group as well as their mothers. The two playpens were located in different rooms during the study.

The design of the experiment is summarized in Table 1. When members of the *S* group reached a mean age of 180 days, the following separation procedure was initiated: first, the homecage

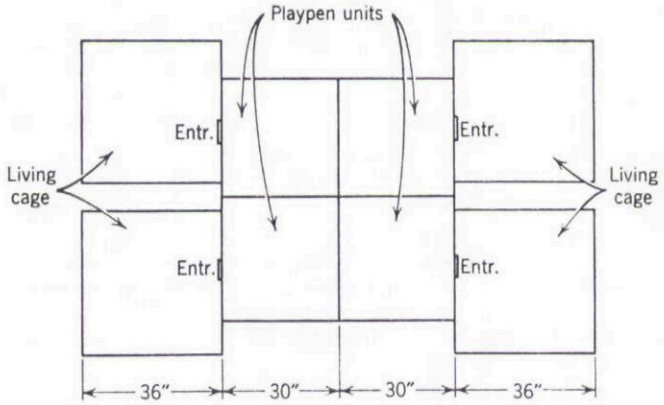


FIG. 1. The playpen apparatus.

TABLE 1. DESIGN OF THE PRESENT EXPERIMENT

Weeks of Age	0-26	27-28	29-30	31-32	33-39	40-46	47-52
Separation Group (S) (4 monkeys)	Mother-peer housing in playpen	Mesh separation from mother (S-M)	Transparent plastic separation from mother (S-P)	Transparent plastic separation from mother and peers (S-I)	Mother-peer housing in playpen	Mother removed; peer housing in playpen	Peer housing in playpen
Control Group (C) (4 monkeys)	Mother-peer housing in playpen	Mother-peer housing in playpen	Mother-peer housing in playpen	Mother-peer housing in playpen	Mother-peer housing in playpen	Mother-peer housing in playpen	Mothers removed; peer housing in playpen

openings leading to the play area were covered by 5 cm wire mesh. The experimental subjects were thus physically confined to the play area, separated by these partitions from their mothers who remained in their respective living units. Limited physical contact between mother and infant was permitted by the openings in the mesh which were of sufficient area to allow arms and legs to reach through to the other side. This condition, termed maternal separation–mesh (*S–M*), was maintained for a period of 14 days. The second stage, maternal separation–transparent plastic (*S–P*), immediately followed. Here, the mesh partitions were removed and replaced with transparent plastic panels, prohibiting all physical contact between mothers and infants. Like the first separation, this stage lasted a period of 14 days. Finally, the infant monkeys were physically separated from one another by insertion of transparent panels into the slots in the play area, confining each subject to the 76 × 76 cm quadrant adjacent to the living cage housing its mother. This period of mother and infant separation (*S–I*) was also 14 days long. All partitions were then removed, reuniting the subjects with their mothers and peers. Members of the C group were not subjected to separations of any form. Rather, they continued to interact with their mothers and fellow-group members.

When the mean age of the S group was 280 days, their mothers were removed from the apparatuses and housed in another part of the laboratory. Mothers were also removed from the control subjects' playpen unit, but due to space considerations within the laboratory at that time, this did not take place until their mean age was 330 days. Subjects in both groups continued to live in the playpen apparatuses until 360 days of age, at which point the study was terminated.

Data collection and analysis

Each subject was observed for a 5-min period 5 days per week, beginning at 12 weeks of age and continuing throughout the remainder of the study. During every observation period, durations of

subject behavior falling into each of 10 predefined behavioral categories (duration categories) were recorded by one of two experienced observers, each of whom had previously been trained to a rigorous laboratory criterion of reliability. Behaviors encompassed by the categories and their operational definitions are listed in Table 2. In addition, a second set of behaviors was recorded during observations of *S* subjects. For these categories, also listed and defined in Table 2, presence or absence during each of the 20 15-sec intervals which comprised each 5-min observation period was recorded (modified frequency categories). All observation periods took place between the hours of 11:00 a.m. and 2:00 p.m. on each week day.

TABLE 2. OPERATIONAL DEFINITIONS OF BEHAVIORAL CATEGORIES

Duration Categories
Mother: ventral–ventral contact between mother and infant, or infant sitting within the embracing arms of the mother.
Mom huddle: any other physical contact with infant's own mother, or orientation toward mother within one body length.
Other mother: any physical contact with or orientation within one body length toward another adult female by infant.
Peer huddle: ventral–ventral or ventral–dorsal contact with another infant (the infant being scored on this behavior would have to have his ventral surface to the other infant).
Peer explore: tactile or oral manipulation of another infant.
Peer play: brief contact play, rough-and-tumble, or pursuit-avoidance play.
Infant reject: negative behavior shown toward another infant such as threat, aggression, or fear grimace.
Environment: tactile or oral manipulation of the cage.
Disturbance mobile: distress vocalizations accompanied by locomotion and/or stereotypy (repetition of a locomotive pattern for three or more cycles).
Disturbance immobile: distress vocalizations accompanying self-clasping and/or rocking and huddling.
Modified Frequency Categories
Screech: High intensity vocalization resembling a scream.
Coo: Drawn-out dove-like cry in a variety of pitches.
Fear Grimace: Facial expression resembling a "smile" produced by the lips being pulled back and exposing the teeth.

The data were analyzed in two stages. First, for each duration category, subject scores were summarized over 2-week blocks and means calculated. Nine of these 2-week blocks were then examined: the 2-week period prior to the mesh separation (24–25 weeks of life), the period of mesh maternal separation (26–27 weeks), the period of transparent plastic maternal separation (28–29 weeks), the period of infant–infant separation (30–31 weeks), the three 2-week periods immediately following reunion (32–33, 34–35, and 36–37 weeks) the 2-week period immediately following removal of the subjects' mothers (40–41 weeks for *S* subjects, 47–48 weeks for *C* subjects), and the final 2-week period of the study (51–52 weeks). For each category, means for each subject for each time period were subjected to a two-way repeated-measures analysis of variance, with subject group (separation vs control) as the independent variable and time period as the repeated measure. For those categories whose analyses yielded significant time period effects and/or significant interactions, Duncan New Multiple Range tests were applied to the appropriate means. Statistical significance was arbitrarily defined as $p < 0{\cdot}05$.

The second stage of data analysis examined more closely the reactions of the *S* subjects to the various separations. For each duration category, scores from each of the *S* subjects were summed over six time periods: the first 2 and the last 3 days of each of the three experimental separations (*S–M*. *S–P*, and *S–I*). In addition, comparable scores were summed over the same time periods for each of the modified frequency categories. Then, for every category, means per session for every subject were calculated for each time period, and these means were subjected to two-way repeated-measures

analyses of variance, with separation period (first 2 days vs last 3 days of separation) and form of separation (*S–M* vs *S–P* vs *S–I*) as repeated measures. For those categories whose analyses yielded significant separation effects and/or interactions, Duncan tests were subsequently applied to the appropriate means.

RESULTS

The results of this study indicated that the experimental subjects' ongoing behavior patterns were grossly affected by the various maternal and infant separations. In contrast to control monkeys who exhibited gradual monotonic changes in behavioral levels over time, members of the separation group exhibited sharp changes in levels of most behaviors following the initial mesh separation which were exaggerated in the subsequent *S–P* and *S–I* separations. Within each separation period there were clear indices of "protest" and "despair" reactions. Levels of most positive behaviors returned to control levels soon after reunion with mother. Finally, both separation and control subjects exhibited major and similar changes in levels of behavior following the permanent removal of their mothers near the end of the study.

Figures 2 and 3 illustrate the mean scores of the duration categories for separation and control subjects during each of the 2-week periods examined in the statistical analyses. The results of these analyses of variance and their accompanying Duncan tests are presented in Table 3. During the initial baseline period, the two groups of subjects did not differ significantly on any measure taken, not a surprising finding since both groups to this point had been reared similarly. During the mesh separation, however, clear group differences emerged in several behavioral categories. *Mother* behaviors were higher for separated subjects than for control subjects, even though their contact was limited to interaction through wire mesh. In contrast, separated

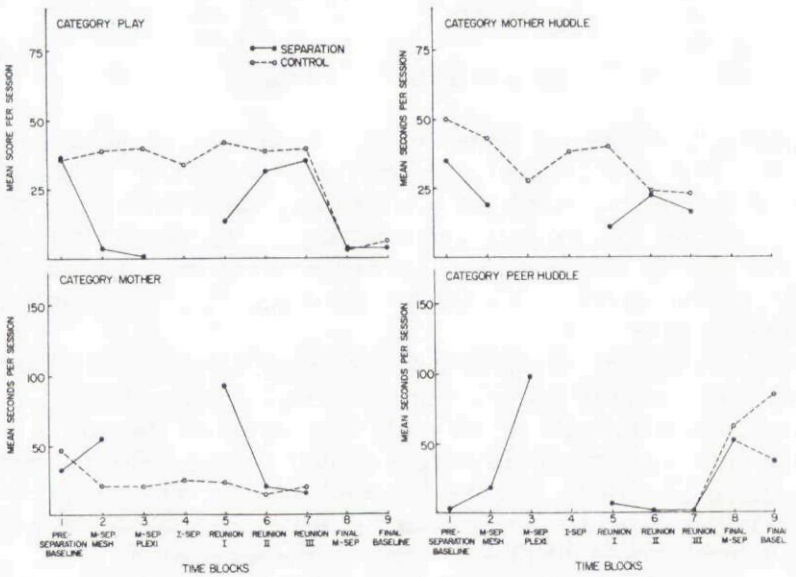


FIG. 2. Mean levels of behaviors.

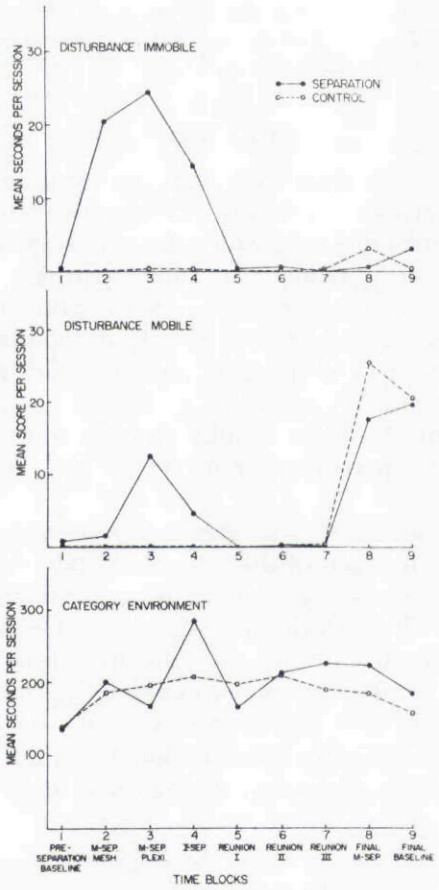


FIG. 3. Mean levels of behaviors.

subjects' levels of *play* dropped almost to zero, significantly below not only control levels but also their own preseparation baselines. In addition, *disturbance immobile* behaviors rose significantly over baseline levels in separated but not control subjects. No other significant differences between separated and control subjects during this time period were disclosed by the analyses. Across both groups of subjects, *environment* behaviors increased over baseline levels, a finding consistent with normal patterns of maturation.

Alteration of behavior was significantly greater for the separated subjects during the period of transparent plastic separation from their mothers. Here, the behaviors of *mother, mom huddle* and *other mother* could not take place due to the partitions. Play, which could in theory occur among separated subjects, failed to appear at all, while remaining at high levels among control subjects. The separated subjects' mean scores in this period for *peer huddle* and *disturbance immobile* were the highest for either group during any of the time periods over the entire study.

Contrary to expectations, experimental subjects did not exhibit qualitatively more severe reactions to the *S–I* separations than they did to maternal separation

TABLE 3. RESULTS OF OVERALL ANOVA'S AND DUNCAN TESTS

Category	Group[a]	Time Block[b]	G × T[b]	Duncan Tests
Mother	0·43	8·09***	4·72***	5 > 1, 2 > 3, 4, 6, 7, 8, 9; *S*5 > all other means; *S*2 > *C*2–9, *S*3–9; *C*1 > *C*7–9, *S*3, 4, 8, 9
Mom huddle	1·23	5·65***	1·31	1 > 3, 4, 5, 7, 8, 9; 2, 5, 6, 7 > 8, 9
Other mother	1·22	1·49	0·78	No Duncan tests
Infant reject	1·67	4·02**	1·50	8, 9 > 1–7
Peer explore	1·16	2·10	0·71	No Duncan tests
Peer huddle	1·38	15·43***	8·00***	3, 8, 9 > 1, 2, 4, 5, 6, 7; *S*3 > *S*1, 2, 4–9, *C*1–8; *C*9 > *S*1, 2, 4–9, *C*1–7; *C*8 > *S*1, 2, 4–7, *C*1–7; *S*8 > *S*1, 2, 4–7, *C*1–7; *S*9 > *S*1, 4, 6, 7, *C*1–7
Play	5·95*	10·95***	4·81***	*C* > *S*; 1, 6, 7, > 2, 3, 4, 8, 9; 2, 3, 5 > 8, 9; *C*1–7, *S*1, 6, 7, > *C*8, 9, *S*2, 3, 4, 8, 9
Environment	0·83	7·09***	2·19**	4 > 2, 3, 5, 6, 7, 8, 9 > 1; 6 > 9; *S*4 > all other means; *C*1, *S*1 > *C*2–8, *S*2, 6–9
Dist. mobile	0·90	3·26**	0·30	8, 9 > 1–7
Dist. immobile	844·15***	38·52***	41·86***	*S* > *C*; 2, 3 > 4 > 1, 5–9; *S*3 > *S*2 > *S*4> *C*1–9, *S*1, 5–9

Tables numbers are *F*-scores. Abbreviations for Duncan tests: *S* = Separated, *C* = Control, 1–9 = respective time blocks. [a] $d.f. = 1/6$, [b] $d.f. = 8/48$, * $p < 0{\cdot}05$, ** $P < 0{\cdot}01$, *** $p < 0{\cdot}001$.

by itself. *Play* and *peer huddle* could not occur, but were more than replaced by significantly elevated levels of environmentally directed behavior. *Disturbance immobile* was higher than during baseline and postseparation time periods but slipped significantly below levels recorded during the prior two separation time periods.

Closer examination of the time course of behavioral change during each of the three separations yielded interesting findings. These data are graphed in Figs. 4 and 5, and the results of the statistical analyses are summarized in Table 4.

First, it is clear that discernible protest–despair sequences of reaction accompanied each form of separation, particularly the latter two. *Disturbance mobile,* a measure of active protest, was significantly greater during the first 2 days of the separations than it was during the last 3 days of each separation. Also, levels of *coo* and *screech* were elevated during the first 2 days of each separation, while fear grimacing was greater for the first 2 days of the *S–M* and *S–I* separations. Contrastingly, behaviors indicative of despair activity associated with depression, *disturbance immobile,* were far more evident during the last 3 days of the separations than they were during the first 2 days. Furthermore, during the *S–P* and *S–I* separations, *environment* behaviors dropped during the last days of each respective separation, consistent with previous reports of characteristic inactivity during despair reactions to separation in monkey subjects.

Second, comparisons across separations indicated that reactions to the *S–M* separation were milder than those to the *S–P* and *S–I* separations. In particular, *environment* and *play* behaviors increased during the *S–M* separation, and both disturbance categories occurred at lower levels during the first 2 and last 3 days of the

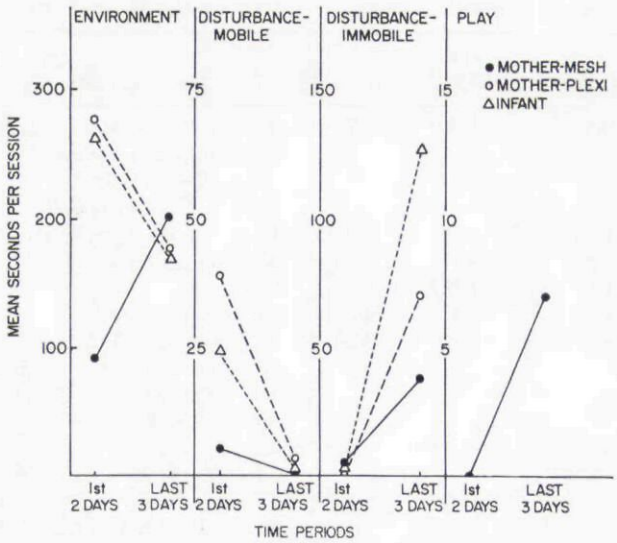


FIG. 4. Mean levels of duration behaviors, separation periods.

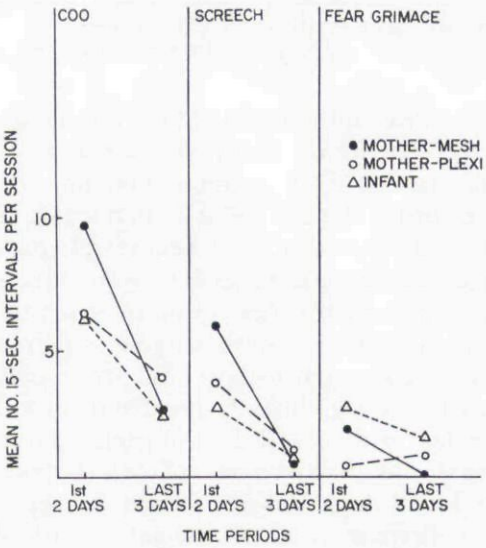


FIG. 5. Mean levels of modified frequency behaviors, separation periods.

S–M separation than during the latter two separations. Differences in severity between the *S–P* and *S–I* separations were less clear-cut. For example, the *S–P* separation yielded higher levels of *disturbance mobile,* while the *S–I* separation yielded higher levels of *disturbance immobile* behaviors.

During reunion periods levels of behavior for *S* subjects returned to those of *C* monkeys at differential rates. For *mom huddle, other mother, peer explore, peer huddle,* and both disturbance categories, separation and control levels did not differ significantly

TABLE 4. RESULTS OF SEPARATION ANOVA'S AND DUNCAN TESTS

Category	Separation[a]	Time[b]	S × T[a]	Duncan Tests
Peer explore	0·30	1·33	1·10	No Duncan tests
Peer huddle	6·22	6·87	2·90	No Duncan tests
Play	11·76*	11·76*	11·76*	1 > 2, 3; B > A; 1B > all other means
Environment	3·99	2·59	12·33**	2A, 3A > 1A
Dist. mobile	2·45	15·04*	2·60	A > B (2A > 1B, 2B, 3B)
Dist. immobile	14·69**	68·78**	10·94**	3 > 1; B > A; 3B > all other means; 3B > 1A, 2A, 3A
Coo	0·64	51·29**	4·05	A > B
Screech	0·97	34·55**	4·05	A > B
Fear grimace	2·50	5·61	7·91*	3A > 1B, 2A, 2B; 1A > 1B

Tables numbers are *F*-scores. Abbreviations for Duncan tests: 1 = *S–M* Separation, 2 = *S–P* Separation, 3 = *S–I* Separation; A = first days of Separation, B = last 3 days of Separation. [a] $d.f. = 2/6$, [b] $d.f. = 1/3$, * $p < 0{\cdot}05$, ** $p < 0{\cdot}01$.

during any of the three reunion time periods. Contrastingly, *play* and *environment* were depressed for the separated subjects until the second reunion period. Also, immediately following reunion, the level of *mother* behavior in the separated subjects was the highest for either group during any time period throughout the study. By the third 2-week period after reunion no significant differences between separated and control subjects in levels of any behavioral category were disclosed by the analyses.

Finally, both groups of subjects exhibited similar reactions to the permanent removal of their mothers from the apparatus, even though the time of removal differed between the two groups by 50 chronological days. Both sets of subjects showed significant increases over previous levels of *infant reject, peer huddle* and *disturbance mobile,* as well as a significant decrease in *play* behavior. These levels were generally maintained until the end of the study. No differences between groups achieved statistical significance, with one exception. Control group levels of *peer huddle* were significantly higher than those of the separated subjects during the final 2-week period of data collection.

DISCUSSION

The results of this study were relatively consistent with predictions: experimental subjects exhibited discernible reactions to each of the three separations, the reactions generally followed the course of protest–despair, and most behaviors returned to control levels following reunion. However, differences among the separations in severity of reaction were not as clear-cut as had been expected, nor were any major differences in reaction to final maternal separation disclosed between separated and control monkeys. These findings merit some discussion.

Of the three separations, the *S–M* clearly produced the mildest reaction among the experimental subjects, as was predicted, although it yielded patterns of disturbance behaviors consistent with the protest–despair sequence found in previous mother–infant separation studies. Specifically, active disturbance was highest immediately following separation, whereas *disturbance immobile* behaviors reached their peak levels near the end of the separation period. However, both *environment*

and *play* behaviors increased during the period of mesh separation, contrary to "typical" despair found in previous studies of young monkeys. We believe the basis for these results lay in the data from the *mother* category of behavior. During the mesh separation the experimental subjects' level of this category rose not only over that of control subjects but also over their own preseparation baseline, even though mother-directed behaviors had to take place through mesh partitions. Much previous work has demonstrated that opportunity for contact with the mother promotes infant activities such as exploration and play (Harlow and Harlow, 1965).

The importance of physical contact with mother for other infant activities was underscored by the data obtained when all physical contact with mother was denied. The infants again exhibited clear protest followed by even clearer despair. Both *disturbance mobile* and *disturbance immobile* behaviors reached their highest 2-week mean levels, while *environment* dropped to its lowest level and play disappeared. Such behavioral patterns are consistent with most previous reports of infant activity following maternal separation in rhesus monkeys, even though the present subjects had been partially limited in their access to mothers 2 weeks prior.

The protest–despair sequence of reaction was repeated during the 2-week period of *S–I* separation. However, although all physical contact with mothers and infants was prohibited, the experimental subjects did not show qualitatively more severe reactions. One explanation for this finding may be that by the time of the *S–I* separation the infants had been exposed to various social restrictions for a 1-month period. Previous work has indicated that some infants tend to recover from effects of maternal separation well within a period of a month (Kaufman and Rosenblum, 1967); it may be that the experimental subjects had adjusted somewhat to the condition of separation from attachment object by the time the *S–I* procedure was initiated.

Nevertheless, clear protest–despair reactions were recorded during *S–I* separation, just as they had been during the two previous separations. As was pointed out previously, not all social separations in primate subjects yield depressive-like despair activity. However, in cases where prior bonds seem strongest, a protest–despair sequence is most likely to occur, independent of the type of attachment bond—mother, inanimate surrogate, or peer—severed by the separation manipulation. This is true of rhesus and pigtail macaques, chimpanzees (van Lawick-Goodall, 1973) and human infants, and conceivably could be true for species such as bonnet macaques were appropriate bonds developed and severed (Harlow and Suomi, 1974).

The removal of mothers near the end of the study produced significant changes in levels of behaviors for both groups of subjects, although the patterns were not identical to those exhibited by experimental subjects following the three earlier separations. Specifically, the monkeys exhibited protest but little or no despair in reaction to permanent maternal separation, although play was depressed and both passive peer contact and aggression increased significantly during both postseparation time periods. This form of reaction to separation was strikingly similar to that reported to result from peer separation of rhesus monkey adolescents (McKinney, Suomi and Harlow, 1972; Bowden and McKinney, 1972) and is somewhat predictable in light of normative development data collected on mother–infant pairs.

Normally, the initially intense rhesus mother–infant attachment bond wanes considerably during the last fourth of the first year of life (Harlow and Harlow, 1965; Hansen, 1966). If the present subjects had relatively normal relationships with their mothers and the data suggest that they did, particularly the control subjects, then the relatively mild reactions exhibited by them are consistent with these normative data.

The finding that separation and control groups showed essentially identical reactions to removal of mother is somewhat at odds with data previously collected at this laboratory. Young *et al.* (1973) reported that separations early in life tended to exaggerate reactions to similar separations later in life. On this basis one might predict that the present experimental monkeys would show more severe responses to removal of mother than control monkeys as a result of their earlier separations. However, the data did not support such a prediction. The fact that Young *et al.*'s monkeys were older than the present subjects and were exposed to brief, repetitive peer separations rather than having their mother removed permanently may account for the different findings across the two studies.

It must be pointed out that certain deficiencies in the experimental design necessitate some restraint in interpretation of the present data. Specifically, limitations on number of subjects available permitted only one control group of four monkeys; ideally, appropriate controls might enable one to determine the precise contribution of each form of separation to the overall constellation of experimental subjects' reactions. In addition, the fact that the mothers of separated and control subjects were removed at different points chronologically, clouds interpretations of the findings that their reactions, despite differential early experiences, were essentially identical.

SUMMARY

Rhesus monkeys reared with mothers and peers for the first 6 months of life were subjected to three consecutive 2-week periods of social separations, progressively increasing the degree of social isolation for the infants. During each period of social separation—mesh permitting physical contact with mothers, physical but not visual separation from mothers but unlimited access to peers, and finally physical separation from both mothers and peers—clear patterns of protest and despair were recorded. Reactions to mesh separations were mild relative to the latter two separations, both equally debilitating. Following reunion, separated subjects returned to behavior patterns similar to those of nonseparated control monkeys. Later in the study mothers were removed permanently from the homecages of all subjects, and all subjects reacted with protest and cessation of play activity. However, there was little evidence of despair and previously separated and control monkeys reacted in similar form despite their different early histories.

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